

Yasin Alperen Bozkurt

Computer Engineer

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EDUCATION

2026 - Present Computer Engineering (M.Sc., GPA: 4.00/4.00) – **Kırıkkale University**
2015 - 2019 Computer Engineering (B.Sc., GPA: 3.01/4.00) – **Fırat University**
2011 - 2014 Information Technology – **Ulus Technical Vocational High School**

SUMMARY

Computer Engineer with 6+ years of experience, specializing in cybersecurity, e-commerce, payment systems, and artificial intelligence. Early in my career, I built PriviaHub, Turkey's first domestic Cyber Range platform; at VNGRS, I was part of the backend team behind Vidyodan, later acquired by Getir, where I managed the payment module. I currently focus on AI-driven development, building an MCP Gateway Proxy and browser-based PII detection. I'm also pursuing graduate research on LLM-based vulnerability detection and false-positive reduction.

RESEARCH & PUBLICATIONS

Just-in-Time Vulnerability Detection in Dynamic Web Frameworks via Plain LLMs

Unpublished Work / Master's Thesis Research

- Designed and evaluated Erbab, a lightweight Just-in-Time (JIT) security-auditing framework that detects vulnerabilities directly from Pull Request git diffs.
- Developed a role-based contextual prompting architecture to reduce the high false-positive rate common in traditional rule-based SAST tools on dynamic web frameworks.
- Applied a pairwise evaluation protocol over vul-intro/vul-fix commit pairs across major OWASP categories (SQLi, XSS, Mass Assignment, Insecure Deserialization) to validate semantic-reasoning accuracy.

Deep Learning Methods for Encrypted Traffic Analysis and Classification in TLS & QUIC Protocols

Graduate Literature Review / Technical Report

- Conducted a systematic literature review of state-of-the-art deep learning architectures (ET-BERT, MIETT, MalMoE) across flow-, packet-, and byte-level traffic representations for TLS 1.3 and QUIC.
- Evaluated core limitations in encrypted traffic classification, including concept drift, zero-day traffic, inference latency, and adversarial evasion attacks (DeepFool, UAP).
- Synthesized emerging research directions in privacy-preserving federated learning and few-shot meta-learning for real-time edge processing.

WORK EXPERIENCE

DefensX – Software Engineer

07/2025 - Present

- Built a fully in-browser PII detection system; benchmarked 20+ open-source models on license, size, and accuracy, and objectively scored results with a dedicated test dashboard.
- Developed an access-management module for an AI MCP gateway that logs end users' AI tool calls.
- Built an AI support assistant on a Vector DB-based RAG architecture, routing support requests to it and reducing the support team's workload.
- Built an admin AI assistant that lets non-technical managers query the database and generate reports in natural language.

DİA Yazılım – Software Engineer

10/2024 - 07/2025

- Redesigned the payment infrastructure end-to-end, enabling a transition to a modular, secure, and extensible payment flow.
- Established a Test-Driven Development (TDD) culture within the team, significantly increasing test coverage and code quality.
- Designed and built a real-time theme preview infrastructure from scratch using Ruby on Rails, Stimulus.js, Turbo, and RSpec.

VNGRS – Software Engineer

05/2021 - 01/2023

- Was part of the backend team for Vidyodan, a live-stream v-commerce platform that reached over 1 million mobile installs; the product was later acquired by Getir.
- Solved concurrency and database optimization problems in high-traffic scenarios triggered by large flash discounts on limited-stock items, where tens of thousands of users attempted simultaneous purchases, preserving system consistency.
- Owned the payment flow, including payment system integrations; delivered billing and real-time reporting integrations.

Web Services Pros – Software Engineer

10/2020 - 10/2024

US-based, remote – full-time 10/2020-04/2021 and 04/2024-10/2024; hourly/freelance in between

- Actively contributed to the development of the Tfg Loyalty and Create An API projects.
- Built CI/CD pipelines on AWS infrastructure (ECS, S3, RDS) using GitHub Actions, speeding up deployment and maintenance.
- Delivered high-quality, accurate software by applying TDD practices extensively.

Privia Security · Software Team Lead

06/2019 - 10/2020

- Built PriviaHub, Turkey's first domestic Cyber Range Platform, from the ground up; shipped both public and enterprise versions to production, enabling its use for security training, employee assessment, and hiring interviews.
- Built an architecture for API-driven, automated management of virtual machines, firewalls, and isolated network infrastructure using KVM, ESX, Docker, Active Directory, and OpenVPN; added a tournament system for user competitions and security-awareness scoring and reporting.
- Promoted to Software Team Lead in recognition of my technical achievements and architectural contributions.

PROJECTS

Erbab | Code Security

erbab.dev

Ruby on Rails, Docker, GitHub Apps, LLM Integrations, CI/CD Pipeline

An AI-powered DevSecOps platform integrated into CI/CD pipelines that performs static code analysis and autonomous security auditing. Built a remediation pipeline that generates autonomous patches for detected logic errors and vulnerabilities and opens GitHub Pull Requests directly; built on an ephemeral container architecture that retains no data after analysis, to protect code confidentiality.

AVCI

getavci.co

Ruby on Rails 8, PostgreSQL, SolidQueue, Turbo Streams, Stimulus.js, AWS Bedrock (Claude)/LLM, Docker, Tailwind CSS

An autonomous AI Sales Agent platform, built after struggling to find customers while selling erbab.dev, that automates B2B companies' sales process end-to-end. Built an architecture that combines web scraping from a company URL, ideal customer profile segmentation, decision-maker identification, and SMTP-based email discovery and verification into a single 6-step agentic pipeline.

Ruby on Rails, PHP, Docker, wget, Excel, CKEditor, SEP

A phishing simulation platform that automates social-engineering tests. Target websites cloned via wget were converted into a PHP-based application, published in a Docker container, and centrally managed through SEP; user lists were imported and managed from Excel files, emails composed with CKEditor were sent to target users, and all interactions were logged and reported.

SKILLS

Backend & Framework	Ruby, Ruby on Rails, RSpec, Sidekiq / SolidQueue, RESTful API Design, Web-Socket, Payment Integrations
Frontend	JavaScript, React, Stimulus.js, Turbo, HTML5/CSS3, Tailwind CSS
Architecture & Principles	OOP, SOLID, MVC, TDD, Clean Code, Design Patterns
Database & DevOps	PostgreSQL, Redis, Docker, AWS (ECS, S3, RDS), CI/CD, GitHub Actions, Linux, Git
Artificial Intelligence	LLM Integrations (Claude, AWS Bedrock), RAG, Vector DB, Agentic Workflow Design, Prompt Engineering
Security & Virtualization	AppSec / DevSecOps, SAST, KVM, ESX, Active Directory, OpenVPN
Languages	Turkish (Native), English